Blood Alcohol Content Modeling

8BM050 - Systems Biology Models - Assignment 2

Introduction

In 2024, Podéus et al. [1] published a comprehensive mathematical model for blood alcohol content (BAC) in humans. Their model can simulate drinks of different alcohol percentages and volumes, as well as combinations with solid food. Understanding how alcohol is absorbed, distributed, metabolized, and eliminated from the body is important for medical applications, public health policy, and forensic science.

In this assignment, you will implement the Podéus model and use it to investigate how various physiological factors affect blood alcohol concentration over time. You will explore the effects of body composition (weight, height, sex), drinking patterns, and potentially other factors such as food consumption and metabolic variations.

The Podéus Model

Because the description of the original Podéus model is slightly overcomplicated, we have simplified the model description. Essentially, the model below is exactly the same as the original from the paper but we have removed equations that were unneccessary and replaced them with a different mechanism that is more friendly to implement.

Model Overview

The model describes the absorption, distribution, metabolism, and elimination of alcohol (ethanol) in the human body. It consists of 10 state variables that track:

- Stomach contents (volume, ethanol concentration, calories)
- Ethanol in the intestine
- Blood alcohol concentration
- Metabolic byproducts (acetate)
- Long-term markers (phosphatidylethanol, or PEth)

The model incorporates several key physiological processes:

1. **Gastric emptying**: Transfer of stomach contents to the intestine
2. **Absorption**: Movement of ethanol from the intestine into the bloodstream
3. **Metabolism**: Breakdown of ethanol in the liver via two enzymatic pathways
4. **Elimination**: Removal of ethanol and metabolites from the body

State Variables

The model uses the following 10 state variables:

1. V_{stomach} (dl): Volume of liquid in the stomach
2. $\text{kcal}_{\text{liquid}}$ (kcal): Energy content effect of liquid in the stomach
3. $\text{kcal}_{\text{solid}}$ (kcal): Current solid food calories in the stomach
4. $\text{EtOH}_{\text{pool}}$ (mg): Ethanol sequestered in stomach contents
5. c_{stomach} (mg/dl): Ethanol concentration in stomach liquid
6. $m_{\text{intestine}}$ (mg): Mass of ethanol in the intestine
7. c_{blood} (mg/dl): Blood alcohol concentration (BAC)
8. c_{acetate} (mg/dl): Plasma acetate concentration
9. PEth (ng/ml): Phosphatidylethanol concentration (long-term marker)
10. $\text{PEth}_{\text{bound}}$ (ng/ml): Bound PEth (cell membranes)

Model Equations

The differential equations governing the system are:

Stomach Volume

We model the stomach volume dynamics as a simple balance between drinking and gastric emptying:

$$\frac{\mathrm{d}[V_{\text{stomach}}]}{\mathrm{d}t} = \sum_{i=1}^{D} R_{\text{drink},i}(t) - R_{\text{emptying}} \tag{1}$$

where:

- $R_{\text{drink},i}(t)$ is a *model input* representing the volume of drink i consumed per minute (dl/min). This is a time-dependent function that you will define based on the drinking pattern you want to simulate.
- R_{emptying} is the gastric emptying rate

$$R_{\text{emptying}} = V_{\max} \cdot \left(\frac{V_{\text{stomach}} - V_{\text{ss}}}{V_{\text{stomach}} - V_{\text{ss}} + K_m} \right) \cdot \exp\left(-\frac{\text{kcal}_{\text{liquid}}}{k_{\text{kcal}}} \right) \tag{2}$$

where:

- $V_{\text{ss}} = 0.001$ dl is the small steady-state stomach volume
- The first term is a Michaelis-Menten relationship with stomach volume
- The exponential term shows that calories slow gastric emptying

Liquid Calories Effect on Emptying

$$\frac{\mathrm{d}\left[\text{kcal}_{\text{liquid}}\right]}{\mathrm{d}t} = \sum_{i=1}^{D} \left(R_{\text{drink},i}(t) \cdot v_{\text{kcal},i} \right) - k_{\text{kcal,clear}} \cdot \text{kcal}_{\text{liquid}} \tag{3}$$

Where $v_{\text{kcal},i}$ is the calories per dl of drink i.

Solid Food Calories

$$\frac{\mathrm{d}[\text{kcal}_{\text{solid}}]}{\mathrm{d}t} = \sum_{i=1}^{M} R_{\text{kcal,solid},i} \tag{4}$$

where:

$$R_{\text{kcal,solid},i} = \text{kcal}_{\text{solid},i,0} \cdot (-1.88) \cdot 0.010 \cdot (0.010 \cdot \max(0, t - t_i))^{0.86} \cdot \exp\left(-(0.010 \cdot \max(0, t - t_i))^{1.86} \right) \tag{5}$$

This function describes the gradual digestion of solid meal i with initial calories $\text{kcal}_{\text{solid},i,0}$ consumed at time t_i.

Ethanol Pool

$$\frac{\mathrm{d}\left[\text{EtOH}_{\text{pool}}\right]}{\mathrm{d}t} = R_{\text{pool,in}} \cdot V_{\text{stomach}} - R_{\text{pool,out}} \tag{6}$$

where:

$$R_{\text{pool,in}} = k_{\text{pool,in}} \cdot \max\left(0, c_{\text{stomach}} - \frac{\text{EtOH}_{\text{pool}}}{V_{\text{solid}}} \right) \tag{7}$$

$$R_{\text{pool,out}} = k_{\text{pool,out}} \cdot \text{EtOH}_{\text{pool}} \tag{8}$$

$$V_{\text{solid}} = \frac{\max(1.0, \text{kcal}_{\text{solid}})}{400} \tag{9}$$

is the volume occupied by solid food

The ethanol pool represents alcohol that is temporarily sequestered in food/mucus in the stomach.

Stomach Ethanol Concentration

$$\frac{\mathrm{d}[c_{\text{stomach}}]}{\mathrm{d}t} = R_{\text{drink,EtOH}} - R_{\text{pool,in}} + \frac{R_{\text{pool,out}}}{V_{\text{stomach}}} \tag{10}$$

where:

$$R_{\text{drink,EtOH}} = \frac{R_{\text{drink}}}{V_{\text{stomach}}} \cdot (c_{\text{drink}} - c_{\text{stomach}}) \tag{11}$$

and $c_{\text{drink}} = p_{\text{EtOH}} \cdot 789.1$ (mg/dl), where p_{EtOH} is the alcohol percentage and 789.1 is the density of ethanol in mg/ml.

Intestinal Ethanol Mass

$$\frac{\mathrm{d}[m_{\text{intestine}}]}{\mathrm{d}t} = R_{\text{emptying}} \cdot c_{\text{stomach}} - R_3 - R_4 \tag{12}$$

where:
- $R_3 = k_3 \cdot m_{\text{intestine}}$ (absorption into blood)
- $R_4 = k_4 \cdot m_{\text{intestine}}$ (first-pass metabolism/loss)

Blood Alcohol Concentration

$$\frac{\mathrm{d}[c_{\text{blood}}]}{\mathrm{d}t} = \frac{R_3}{V_{\text{blood}}} - R_5 \cdot \left(\frac{V_{\text{liver}}}{V_{\text{blood}}}\right) \tag{13}$$

where:
- V_{blood} is the blood volume (see below for calculation)
- $V_{\text{liver}} = 15.0$ dl is the liver volume
- R_5 is the total hepatic metabolism rate, defined below.

The liver metabolizes ethanol via two pathways:

1. *Alcohol dehydrogenase (ADH)*:

$$V_{\text{ADH}} = V_{\text{max,ADH}} \cdot \frac{c_{\text{blood}}}{K_{\text{m,ADH}} + c_{\text{blood}}} \tag{14}$$

2. *Cytochrome P450 2E1 (CYP2E1)*:

$$V_{\text{CYP2E1}} = V_{\text{max,CYP2E1}} \cdot \frac{c_{\text{blood}}}{K_{\text{m,CYP2E1}} + c_{\text{blood}}} \tag{15}$$

Total metabolism: $R_5 = V_{\text{ADH}} + V_{\text{CYP2E1}}$

The blood volume is calculated based on sex, body height (h in meters), and body weight (w, in kg):

$$V_{\text{blood}} = (a \cdot h^3 + b \cdot w + c) \cdot 10.0 \quad [\text{dl}] \tag{16}$$

where:
- For females: $a = 0.3561, b = 0.03308, c = 0.1833$
- For males: $a = 0.3669, b = 0.03219, c = 0.6041$

Plasma Acetate

$$\frac{\mathrm{d}[c_{\text{acetate}}]}{\mathrm{d}t} = R_5 - R_6 \tag{17}$$

where $R_6 = k_6 \cdot c_{\text{acetate}}$ is the acetate elimination rate.

Phosphatidylethanol (PEth)

$$\frac{\mathrm{d}[\text{PEth}]}{\mathrm{d}t} = R_{\text{PEth}} - R_{\text{PEth,bind}} + R_{\text{PEth,release}} - R_{\text{PEth,clear}} \tag{18}$$

$$\frac{\mathrm{d}[\text{PEth}_{\text{bound}}]}{\mathrm{d}t} = R_{\text{PEth,bind}} - R_{\text{PEth,release}} \tag{19}$$

where:
- $R_{\text{PEth}} = k_{\text{PEth}} \cdot c_{\text{blood}}$ (formation)
- $R_{\text{PEth,clear}} = k_{\text{PEth,out}} \cdot \text{PEth}$ (clearance)
- $R_{\text{PEth,bind}} = k_{\text{PEth,bind}} \cdot \text{PEth}$ (binding to membranes)
- $R_{\text{PEth,release}} = \max\big(0, k_{\text{PEth,release}} \cdot (\text{PEth}_{\text{bound}} - \text{PEth})\big)$ (release from membranes)

Model Parameters

The model uses the following parameter values (from Podéus et al. 2024):

Parameter	Value	Description
k_{PEth}	1.4599×10^4	PEth formation rate
$k_{\text{PEth,out}}$	1.5695×10^2	PEth clearance rate
$k_{\text{PEth,bind}}$	1.7416×10^4	PEth binding rate
$k_{\text{PEth,release}}$	5.8081×10^{-3}	PEth release rate
$k_{\text{pool,in}}$	1.0155×10^1	Ethanol pool sequestration

$k_{\text{pool,out}}$	4.0227×10^{-1}	Ethanol pool release
V_{max}	2.1545×10^{3}	Max gastric emptying rate
K_m	1.7793×10^{4}	Gastric emptying half-saturation
k_{kcal}	1.5843×10^{2}	Calorie effect on emptying
k_3	8.4172×10^{1}	Intestinal absorption rate
k_4	8.3001×10^{2}	First-pass metabolism rate
k_6	1.3079×10^{-1}	Acetate elimination rate
$V_{\text{max,ADH}}$	9.6381×10^{-1}	Max ADH metabolism rate
$V_{\text{max,CYP2E1}}$	1.7148×10^{-1}	Max CYP2E1 metabolism rate
$K_{\text{m,ADH}}$	9.2200	ADH Michaelis constant
$K_{\text{m,CYP2E1}}$	3.6880×10^{1}	CYP2E1 Michaelis constant
$k_{\text{kcal,clear}}$	6.1804×10^{-3}	Calorie clearance rate

Currently Implemented Simulation

You are provided with the following base code folder structure:

- `simulate_podeus.py`: Main script to run simulations and generate plots
- `podeus/`
  - `inputs.py`: Functions to define drink and meal inputs (already implemented)
  - `equations.py`: Functions to calculate rates and derivatives based on the model equations (incomplete, you need to write code to complete the implementation)
  - `__init__.py`: Python file to make the `podeus` folder a package so you can just do `import podeus` (already implemented)

Assignment Task

Running the `simulate_podeus.py` script will currently generate a plot of blood alcohol concentration over time, but it does not yet implement the full Podéus model equations. The current output is just a placeholder to show where the BAC curve will be plotted once you implement the model.

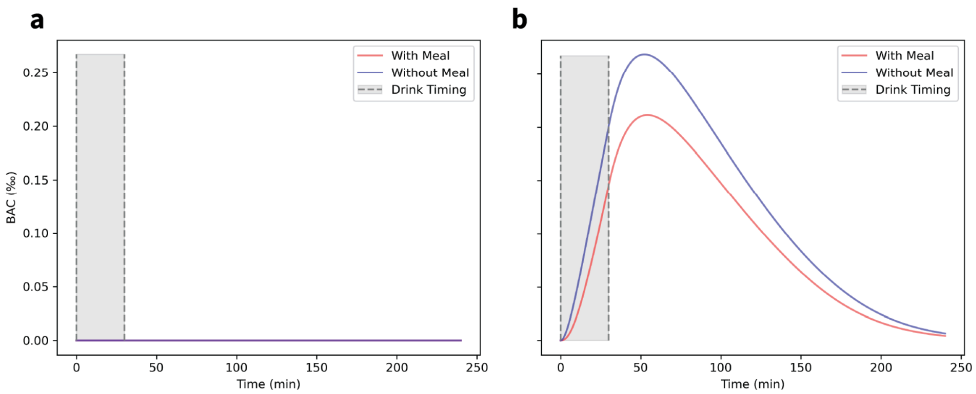

Figure 1: Example BAC curves generated with the Podéus model. **(a)** Curve generated by the provided code with incomplete equations. **(b)** Curve generated by a correct implementation of the full model. The correct implementation shows a realistic BAC curve that rises during drinking and falls afterward, while the incomplete version does not capture these dynamics.

Steps

1. In `podeus/equations.py`, stomach and intestinal ethanol dynamics are already implemented, but the equations in the blood compartment are not yet implemented (set to `0.0` for now). The current script will produce the figure shown in Figure 1a, which does not capture the expected BAC dynamics. Your task is to implement the equations for blood alcohol concentration, plasma acetate, and PEth dynamics based on the model equations described above.
2. Once you have implemented the equations, run the `simulate_podeus.py` script to generate the BAC curve. If you used the settings provided in the script, you'll get the curve in Figure 1b
3. After implementing the model, formulate a research question that you can test with the model.
4. Design a set of computational experiments to investigate your research question.
5. Run the experiments and analyze the results. Generate figures to illustrate your findings.
6. Discuss your results in the context of the model. For example, discuss whether the results are consistent with known physiological principles, and how the model equations and parameters explain the observed dynamics.
7. Write a short report containing your research question, methods, results, and discussion.

Minimum Viable Product (MVP) Requirements (Grade: 7.0)

To help you get started, we define a minimum viable product (MVP), containing a list of questions that you should address in your report. If you choose to implement the MVP, you will receive a maximum grade of 7.0. Higher grades can be achieved by implementing further extensions or a completely different task that you think of yourself.

To clarify, you are **NOT** required to do the MVP. You can choose to do a different task or implement extensions instead. The MVP is just a suggested direction for this assignment.

Important note: An alternative task that you think of yourself is not guaranteed to get you a higher grade than a 7.0, so make sure to design a task that is sufficiently interesting and challenging if you choose to go that route.

MVP Questions to Answer

Think of a research question related to how body composition influences BAC dynamics. In you experiments, answer the following questions:

- What differences arise in BAC curves for different body compositions?
- How does sex influence these dynamics?
- How do these differences relate these factors are reflected in the model equations and parameters?
- Does the amount of alcohol consumed interact with body composition effects? What about drinking speed?

Ideas for Alternative or Extended Tasks

These are just some ideas for alternative tasks if you want to go beyond the MVP. These are intentionally vague. Be creative and design your own experiments to investigate these questions. You're encouraged to also think of your own interesting questions or extensions beyond these suggestions.

1. **Enzyme Deficiencies**
   - Are there known genetic deficiencies affecting enzyme activity?
   - Extend enzymatic breakdown or add new pathways to simulate these conditions
2. **Liver Disease**
   - Model cirrhosis or hepatitis
   - Investigate how reduced liver function affects BAC dynamics
   - Calculate how much longer alcohol persists in the system
3. **Tool Development**
   - Create a user-friendly interface for simulating different drinking scenarios
   - Allow users to input personal parameters and visualize BAC curves
   - Compare the model predictions with other tools available online

Deliverables

Submit the following:

1. **Python code**: Your implementation of the Podéus model
   - Well-commented and organized
   - Include all functions for running experiments
   - Reproducible: running your code should regenerate all results
2. **Report (PDF)**:
   - For MVP: around 2 pages (excluding title page and references), absolute maximum 4 pages
   - With extensions: maximum 8 pages
   - Include: introduction, methods, results (with figures), discussion, conclusion
   - References in proper format
   - Please, no appendices.
3. **Figures**: All plots should be:
   - Clear and readable (proper font sizes, labels)
   - Have descriptive captions
   - Use consistent styling
   - Include legends when showing multiple curves

Tips for Success

- **Start early**: Implementing the model correctly already takes time
- **Test incrementally**: Don't implement all equations at once; test each part
- **Use version control**: Save working versions as you progress
- **Sanity checks**: Does BAC increase during drinking and decrease after? Do values make physical sense?
- **Plot intermediate variables**: Visualizing stomach volume, intestinal ethanol, etc. helps debug
- **Physical units**: Pay attention to unit conversions (dl vs. ml, min vs. hours, percentages vs. fractions)
- **Document assumptions**: If you modify the model, clearly explain why

Grading Rubric

Component	Points	0%	20%	40%	60%	80%	100%
Model Completion	10	Absent or incorrect	Partial implementation	Most equations implemented	All equations implemented with minor errors	Correct implementation with good structure	Correct and efficient implementation
Design of experiments	15	Lacking or inappropriate	Basic design with limited scope	Adequate design addressing some questions	Well-designed experiments addressing key questions	Comprehensive design with multiple factors	Complete and thorough experimental design
Quality of results	20	Poor or missing	Basic results with limited analysis	Adequate results with some analysis	Good results with clear analysis	Comprehensive results with insightful analysis	Extensive results with deep insights
Discussion and conclusion	20	Missing or irrelevant	Basic discussion with limited insight	Adequate discussion addressing some points	Good discussion with clear insights	Comprehensive discussion with strong insights	Complete and thoughtful discussion
Extended explorations (grade based on results and discussion/ conclusion)	30	None, or at the same level of the MVP	Slightly more extensive than MVP. Has some interesting insights, but leaves some gaps in the analysis.	Slightly more extensive than MVP. Some interesting insights. Good analysis, with few gaps	Definitely more extensive than MVP. Interesting insights. Good analysis with few gaps.	Definitely more extensive than MVP. Interesting insights. Good analysis.	The experiments are creative, interesting, and go well beyond the MVP. The insights are very interesting and the analysis is thorough.
Report and code style	5	Poorly organized or hard to read	Basic organization with some clarity	Adequate organization and readability	Well-organized and clear	Very well-organized with good readability	Exceptionally clear and well-structured

Maximum total points: 100. Grade is calculated as (total points / 10).

Questions?

If you have questions about the assignment:

- Check the discussion forum first
- Ask on the forum
- Attend BZ
- Private questions; email the course staff

Good luck!

Bibliography

[1] H. Podéus *et al.*, “A physiologically-based digital twin for alcohol consumption—predicting real-life drinking responses and long-term plasma PEth,” *npj Digital Medicine*, vol. 7, no. 1, p. 112, May 2024, doi: 10.1038/s41746-024-01089-6.